Supplementary Material: Independent COMSOL Long-Window Follow-Up Experiment

Purpose and Scope

This supplementary note reports an independent COMSOL follow-up experiment performed in response to the editor's concern that the original public-benchmark setting used a five-second observation window followed by a two-second extrapolation window. The purpose is to examine whether the proposed MI-PINN framework still mitigates error growth when the observation-free interval is extended.

The experiment is deliberately reported as an *independent COMSOL follow-up*, not as a strict extension of the public Dou et al. benchmark data. The original manuscript results over the public benchmark remain defined by the selected 0–5 s observation window and 5–7 s extrapolation window. The present supplementary experiment uses newly generated COMSOL reference sequences and therefore should not be merged numerically with the public-benchmark RMSE or R^2 values.

Data and Sampling Protocol

Two independent COMSOL 6.3 cases were used: a single-cylinder wake at $Re = 100$ and a three-cylinder wake at $Re = 100$. For both cases, the available time interval was 60–75 s with an output interval of $\Delta t = 0.05$ s. The 60–65 s segment was used as the observation window, and 65–67 s, 65–70 s, and 65–75 s were used as progressively longer observation-free evaluation windows.

The spatial regions were matched to the selected learning windows used in the manuscript:

- Single-cylinder case: $[1, 9] \times [-2, 2]$.
- Three-cylinder case: $[6, 14] \times [-2, 2]$.

The exported COMSOL fields were regridded to a spacing of $\Delta x = \Delta y = 0.05$, giving $161 \times 81 = 13041$ spatial points for each case. Sparse observations were generated from $N_a = 2000$ fixed Eulerian virtual probes selected within the corresponding region. Velocity labels were used only in the observation window. During the extrapolation window, velocity observations were masked; the models were evaluated against the reserved COMSOL reference fields after training.

Because this COMSOL follow-up uses $\Delta t = 0.05$ s, a two-second direct prediction horizon corresponds to 40 time steps. This differs from the public benchmark setting in the manuscript, where $\Delta t = 0.01$ s and a two-second horizon corresponds to 200 time steps. This difference is one reason why the follow-up experiment is reported separately from the main public-benchmark results.

Quantitative Results

Table 2 reports speed-field R^2 and RMSE over the three progressively longer extrapolation windows. MI-PINN consistently reduces RMSE relative to the standard PINN baseline in both the single-cylinder and three-cylinder cases.

Table 1: Training and evaluation protocol for the independent COMSOL follow-up experiment.

Item	Setting
Observation window	60–65 s
Evaluation windows	65–67 s, 65–70 s, 65–75 s
Output time step	$\Delta t = 0.05$ s
Virtual probes	$N_a = 2000$ fixed Eulerian probe locations
Observation mini-batch	$N_{\text{obs}} = 4000$ samples per iteration
Physics collocation mini-batch	$N_f = 20000$ points per iteration
Mamba consistency mini-batch	$N_m = 1140$ extrapolation-window samples
PINN optimization	20000 Adam iterations
Mamba optimization in MI-PINN	3000 Adam iterations

Table 2: Independent COMSOL long-window follow-up results.

Case	Method	R^2 65–67 s	RMSE 65–67 s	R^2 65–70 s	RMSE 65–70 s	R^2 65–75 s	RMSE 65–75 s
Single	PINN	0.835830	0.097489	0.266361	0.205805	-0.962353	0.335503
Single	MI-PINN	0.982278	0.032031	0.982181	0.032074	0.970701	0.040995
Three	PINN	0.775118	0.095083	-1.000723	0.277456	-2.783403	0.390878
Three	MI-PINN	0.902198	0.062704	0.822908	0.082547	0.734057	0.103632

Table 3: RMSE reduction of MI-PINN relative to PINN in the independent COMSOL follow-up experiment.

Case	65–67 s	65–70 s	65–75 s
Single	67.14%	84.42%	87.78%
Three	34.05%	70.25%	73.49%

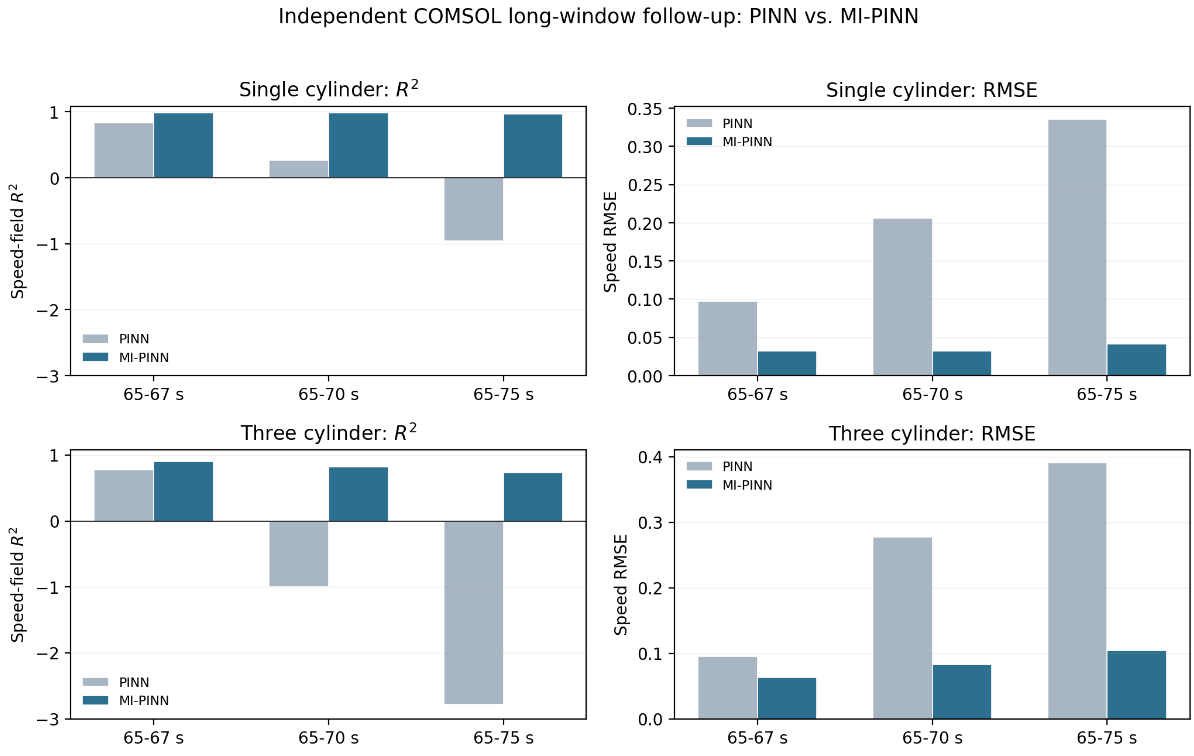

Figure 1: Independent COMSOL long-window follow-up experiment. The standard PINN baseline shows clear error growth as the observation-free interval is extended, whereas MI-PINN retains higher speed-field R^2 and lower RMSE in both cylinder-wake cases.

Interpretation

The follow-up experiment supports a limited but useful conclusion. Under the independent COMSOL setting, the standard PINN baseline deteriorates substantially as the observation-free window is extended from 2 s to 5 s and 10 s. In contrast, MI-PINN maintains higher field-level accuracy, indicating that the Mamba-based temporal-prior consistency term helps reduce extrapolation error growth in the tested longer-window setting.

This evidence does not prove arbitrary long-term forecasting ability, nor does it convert the original public benchmark into a long-horizon benchmark. It instead provides an additional, traceable, independent-COMSOL result showing that the proposed temporal-prior mechanism remains beneficial when a longer reference sequence is available.

Traceability

The formal training root is `E:/yzy/eaai_cfd_rebuild/editor2_mipinn_training/formal_runs`. The corresponding run suffixes are:

Case	Method	Run suffix
Single	PINN	`single/pinn/run_20260611_163150`
Single	MI-PINN	`single/mipinn/run_20260611_165753`
Three	PINN	`three/pinn/run_20260611_171830`
Three	MI-PINN	`three/mipinn/run_20260611_173245`